

Ex: $a^{20} \equiv 1 \pmod{21}$ for $a = 1, 8, 13, 20$

but $21 = 3 \cdot 7$ is composite.

Defⁿ We say that n is a pseudoprime to the base $a \in \mathbb{Z}$ if

$$a^{n-1} \equiv 1 \pmod{n}.$$

Exercise: This is only possible if $\gcd(a, n) = 1$.

How likely is the Fermat primality test to be correct?

We now show that if the Fermat primality test can prove that n is composite because it is not a pseudoprime to the base a , then many other bases will also work, assuming that $\gcd(a, n) = 1$.

Propn let n be composite. Suppose that $a \in \mathbb{Z}$

satisfies $a^{n-1} \not\equiv 1 \pmod{n}$ and $\gcd(a, n) = 1$.

Then at least half of the elements x of $\mathbb{Z}_n^*$ satisfy $x^{n-1} \not\equiv 1 \pmod{n}$.

Proof: Define a function

$$f: \mathbb{Z}_n^* \longrightarrow \mathbb{Z}_n^*, \quad x \longmapsto ax \pmod{n}.$$

Since $\gcd(a, n) = 1$, f is well-defined and bijective.

let $X = \{x \in \mathbb{Z}_n^* : x^{n-1} \equiv 1 \pmod{n}\}$

and $Y = \{x \in \mathbb{Z}_n^* : x^{n-1} \not\equiv 1 \pmod{n}\}.$

Claim: $f(X) \subseteq Y$. Indeed, if $x \in X$, then

$$\begin{aligned} f(x)^{n-1} &\equiv (ax)^{n-1} \equiv a^{n-1} x^{n-1} \\ &\equiv a^{n-1} \not\equiv 1 \pmod{n}. \end{aligned}$$

Thus, since f is injective,

$$|X| = |f(X)| \leq |Y|.$$

The claim follows since $|\mathbb{Z}_n^*| = |X| + |Y|$.

Remark: Recall that $\frac{\phi(n)}{n^{1-\delta}} \rightarrow \infty$ for any $\delta > 0$.

Fast exponentiation

Both RSA ("m $\mapsto m^e \bmod n$ ", " $c \mapsto c^d \bmod n$ ") and the Fermat primality test (" $a^{n-1} \bmod n$ ") require raising integers to large powers modulo some number. In practice, we should not compute $a^{n-1} \bmod n$ by first determining a^{n-1} (which can be very large). Instead, we may use the following algorithm.

Input: $a \in \mathbb{Z}, e, n \geq 1$

Output: $a^e \bmod n$

- Write $e = e_0 + 2e_1 + \dots + 2^k e_k$,
where $e_0, \dots, e_k \in \{0, 1\}$ and $e_k = 1$. "binary expansion of e "
- Use the recursive formula

$$a^{2^{j+1}} \bmod n = (a^{2^j} \bmod n)^2 \bmod n$$

to compute $a^{2^j} \bmod n$ for $j = 0, \dots, k$.

- Rebrine $a^e \bmod n = \prod_{j \text{ st. } e_j = 1} a^{2^j} \bmod n$,

again reducing mod n after each multiplication.

Advantages of this method: • A good implementation uses $O(\log e)$ multiplications.

- The largest intermediate integer is $\leq (n-1)^2$.

Fast exponentiation in Python:

`pow(a, e, n)`

Example: Compute $6^{73} \bmod 100$.

Don't use $6^{73} \approx 6.38 \cdot 10^{56}$ directly.

Instead: $73 = 1 + 8 + 64$
 $= 2^0 + 2^3 + 2^6$

Now: $6^0 \equiv 1$, $6^1 \equiv 6$, $6^2 \equiv 36$

$$6^4 \equiv 36^2 \equiv 96, \quad 6^8 \equiv 96^2 \equiv 16,$$

$$6^{16} \equiv 16^2 \equiv 56, \quad 6^{32} \equiv 56^2 \equiv 36,$$

$$6^{64} \equiv 36^2 \equiv 96, \quad \text{all mod } 100$$

$$\text{Hence: } 6^{73} \equiv (6^1 \cdot 6^8) 6^{64}$$

$$\equiv (6 \cdot 16 \bmod 100) 96 \bmod 100$$

$$\equiv (96 \bmod 100)^2 \bmod 100 \equiv 16 \pmod{100}$$

Back to primality testing:

Suppose that n is a composite number.

Then the Fermat primality test can probably detect

that n is composite if there exists just a single

$a \in \mathbb{Z}$ with $\gcd(a, n) = 1$ and $a^{n-1} \not\equiv 1 \pmod{n}$.

(Because there will be many such numbers in $\mathbb{Z}_n^\times$.)

Q: Given a composite n , does there always exist an integer $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$ and $a^{n-1} \not\equiv 1 \pmod{n}$?

Defⁿ A composite (!) number $n \geq 4$ is a Carmichael number if

$$a^{n-1} \equiv 1 \pmod{n}$$

for all $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$.

Facts: • n is a Carmichael number iff

$$a^n \equiv a \pmod{n} \text{ for all } a \in \mathbb{Z}.$$

(assuming, as before, that n is composite!)

- Carmichael numbers are odd, square-free, and divisible by at least three primes.

Ex: The first Carmichael numbers are

561, 1105, 1729, ...



R.D. Carmichael 1910

Thm: (Alford, Granville, Pomerance 1994)

There are infinitely many Carmichael numbers.

In conclusion:

- The Fermat primality test will report that a Carmichael number n is "probably prime" for any choice of a base $a \in \mathbb{Z}_n^\times$.
- If our random integer a from $2, \dots, n-2$ happens to satisfy $\gcd(a, n) > 1$, then the Fermat primality correctly detects that n is composite. This is very unlikely.
- There are too many Carmichael numbers to ignore.
 $\leadsto$ The Fermat primality test is not reliable.

Better primality tests:

- (In practice:) The $\rightarrow$ Miller-Rabin primality test is an efficient, practical, probabilistic algorithm which can detect every composite number with high probability.
- (In theory:) The $\rightarrow$ AKS primality test (2002!) is a deterministic polynomial-time algorithm for testing primality of integers.

Title of the breakthrough paper: "PRIMES is in P."

Prime generation:

To find a random (probable) prime with a certain number of binary digits, pick a random integer of that length and test if it is (probably) prime.

Repeat until it is.

(In practice, a number with sufficiently high probability of being prime is usually good enough.)

The above approach works thanks to the

Prime Number Theorem: (J. Hadamard, C. J. de la Vallée Poussin 1896)

Let $\pi(n) := \# \text{ primes } \leq n$.

Then

$$\pi(n) \sim \frac{n}{\log n}$$

as $n \rightarrow \infty$.

Hence:

primes between n and $2n$

$$\approx \pi(2n) - \pi(n) \approx \frac{2n}{\log(2n)} - \frac{n}{\log n}$$

$$= \log n + \log 2$$

$$\approx \log n$$

$$\approx \frac{n}{\log n}$$

so the probability that a random integer between

$$n \text{ and } 2n \text{ is prime is } \approx \frac{1}{\log n} > 0.$$